National University of Computer and Emerging Sciences

Data Mining (DS3002)

Date: Feb 28th 2024

Course Instructor

Miss Eesha Tur Razia Babar

Sessional-I Exam

Total Time: 1 Hours Total Marks: 25

Total Questions: 04

Semester: Spring-2024

Campus: Lahore

Dept: AI and Data Science

________________________ ________ ________ __________________

Student Name Roll No Section Student Signature

________________________ __________________

Vetted by Vetter Signature

CLO 1: Understand basic concepts of data mining.

Q1: Bayes Classifier and Naïve Bayes Classifier ***[0.5+0.5+1+1+1+1+1.5+1.5+2 = 10 marks]***

x_1	x_2	y
0	0	1
0	0	1
0	1	1
0	0	0
1	1	0
1	0	0
0	1	1
0	1	1

1a) Using the data given in the Table above to calculate the following probabilities required for the naïve bayes classifier.

i) **P (y = 0) ii)**
P (y =1) iii)
P (x1 = 0/y = 0)
iv) **P (x2 = 0/y = 0)**
v) **P (x1 = 0/y = 1)**
vi) **P (x2 = 0/y = 1)**

1b) Write p (y = 0/ x1 = 0, and x2 = 0) using naïve bayes classifier.
1c) Write p (y = 1/ x1 = 0, and x2 = 0) using naïve bayes classifier
1d) Write p (y = 1/ x1 = 0, and x2 = 1) using naïve bayes classifier

(a) Write down the probabilities necessary for a naïve Bayes classifier:

$p(y=1) = 5/8$

$p(x_1=1 \mid y=0) = 2/3$ $\quad$ $p(x_1=1 \mid y=1) = \emptyset$.

$p(x_2=1 \mid y=0) = 1/3$ $\quad$ $p(x_2=1 \mid y=1) = 3/5$.

(b) Using your naïve Bayes model, what value of y is predicted given observation $(x_1, x_2) = (00)$?

Compare: $p(y=0)\, p(x_1=0 \mid y=0)\, p(x_2=0 \mid y=0)$ vs $p(y=1)\, p(x_1=0 \mid y=1)\, p(x_2=0 \mid y=1)$

$3/8 \cdot 1/3 \cdot 2/3$ vs $5/8 \cdot 1 \cdot 2/5$

$= \frac{6}{72}$ vs $1/4$.

$\Rightarrow$ predict $\hat{y} = 1$.

(c) Using your naïve Bayes model, what is the probability $p(y = 1|x_1 = 0, x_2 = 1)$?

Compare: $p(y=0)\, p(x_1=0|y=0)\, p(x_2=1|y=0)$ $\quad$ $p(y=1)\, p(x_1=0|y=1)\, p(x_2=1|y=1)$

$3/8 \cdot 1/3 \cdot 1/3$ $\quad$ $5/8 \quad 1 \quad 3/5$

$= \frac{3}{8 \cdot 9}$ $\quad$ $\frac{3}{8}$

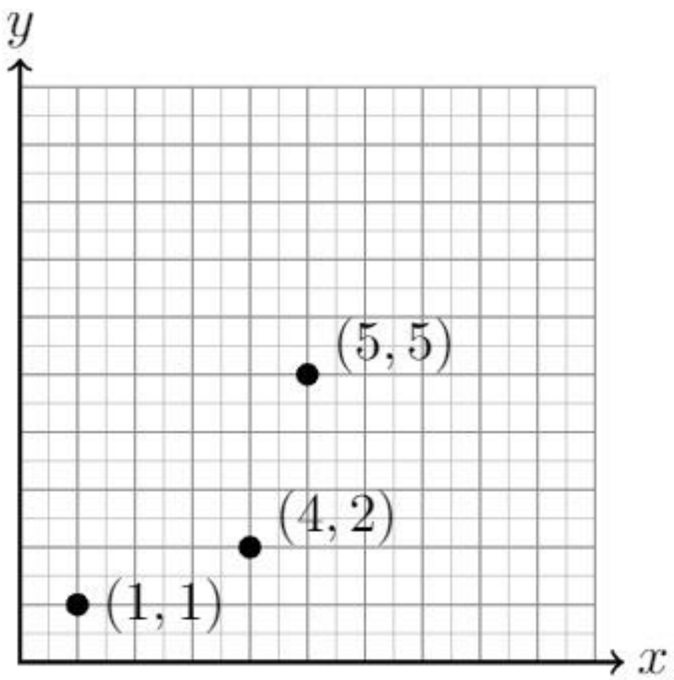
$\rightarrow p(y=1|x=01) = \frac{3/8}{3/8 + 3/72} = \frac{27}{27+3} = \frac{27}{30} = .9$.

CLO #2: Apply data mining techniques to extract information from large data sets

---

Q2: KNN regression ***[2+3 = 5 marks]***

Consider the data points shown below, for a regression problem to predict y given a scalar feature x.



1) ***Compute the training mean squared error (MSE) of a 1-nearest neighbor predictor.***
0

2) Compute the leave-one-out cross-validation error (MSE) of a 1-nearest neighbor predictor. 6.33

CLO #2: Apply data mining techniques to extract information from large data sets

---

Q3: Decision Trees ***[3+3+1 = 7 marks]***

In the following table X1 and X2 are features and Y is class label. Use the data provided in the following table to

X_1	X_2	Y
T	T	T
T	F	T
T	T	T
T	F	T
F	T	T
F	F	F
F	T	F
F	F	F

1) **Compute Information gain for feature X1.**
2) **Compute Information gain for feature X2.**
3) **Report the feature which provides the best split.**

CLO #2: Apply data mining techniques to extract information from the large data sets

---

Q4: True/False [1+1 = 2 marks]

1) **In data mining algorithms we always favor higher variance over higher bias. False**
2) **SVMs are also called minimum margin classifiers. False**
3) **Adaboost classifier is an example of bagging based classifier. False**